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TdP

Python modules and packages

Structuring large(r) projects

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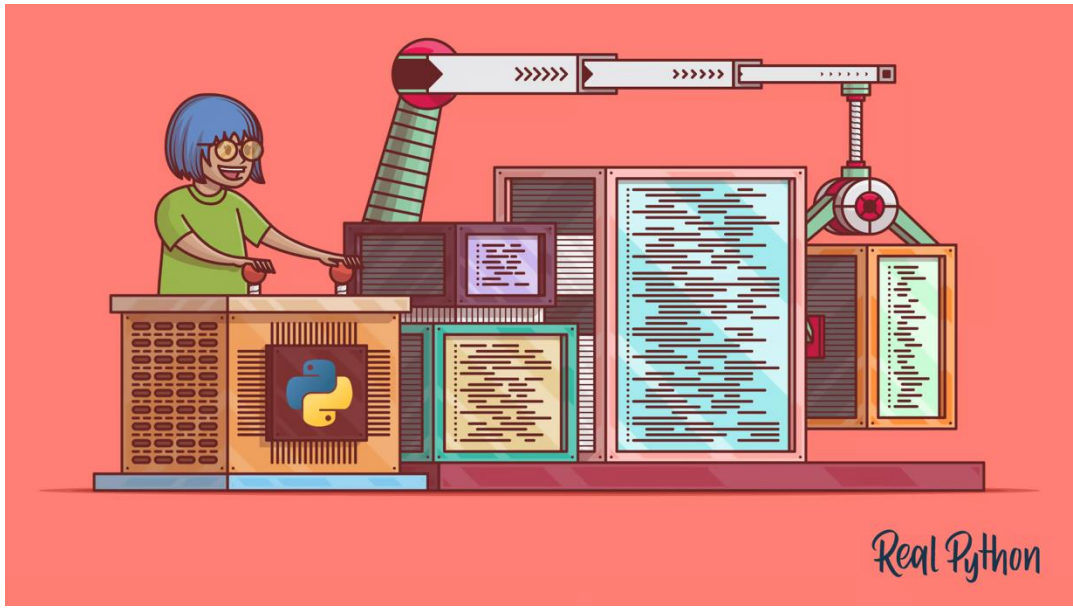

https://realpython.com/python-modules-packages/

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Goal

- Split a large program in multiple files
- Make re-usable library of classes
- Import and use additional libraries

Modules

- Modules are collections of classes, functions, variables, and declarations
- The `import` statement runs the module source and makes the definitions available
- The defined names are created in a separated *namespace* to avoid confusion

Standard library modules

- random, math, csv, operator, os, dataclasses, datetime, … 200 more
- https://docs.python.org/3/library/index.html

User-defined modules

- My python sources
- Files and directories in my project
- Files and directories in the Python search path

Downloaded modules

- From https://pypi.org/ - over 500k projects
- Install using pip

Creating a Python module

- Just create a `.py` file
  - In the same directory of your main file
  - Should contain declarations, only
- The *name of the file* is the name of the module
  - The argument of `import`
- All names defined *at the top-level* become visible properties of the module
  - Constants
  - Functions
  - Classes
  - Variables (bad idea)

voto.py

```
MAX_VOTO = 30

class Voto:
    def __init__(self, esame, cfu, punteggio, lode, data):
        ...
    def __str__(self):
        ...
    def __repr__(self):
        ...

class Libretto:
    def __init__(self):
        ...
    def append(self, voto):
        ...
    def media(self):
        ...


def voto_casuale():
    ...
```

main.py

```
import voto
```

The `import` statement

- `import` *`module_name`*
  - Imports the definitions from `module_name`. They will be accessible as `module_name.definition`
  - Example: `import math` ; use `math.sin(math.pi)`
- `import` *`module_name`* `as` *`alt_name`*
  - Imports the definitions from `module_name`. They will be accessible as `alt_name.definition`
  - Example: `import cmath as c` ; use `c.sqrt(-1)`

The `from…import` statement

- `from module_name import name(s)`
  - Import one or more `name(s)` from `module_name`, and make them available in the current namespace
  - Example: `from math import pi, sin, cos` ; use `sin(pi)`
- `from module_name import name as alt_name`
  - Import one `name` from `module_name`, and make it available in the current namespace as `alt_name`
  - Example: `from cmath import sqrt as csqrt` ; use `csqrt(-1)`
- `from module_name import *`
  - Import all available names from `module_name`, and make them available in the current namespace
  - Except names starting with '_' (underscore)… they are ignored
  - Somewhat dangerous… may have conflicts with local names or other module's names

Querying available names: `dir()`

- The `dir()` function shows the list of names defined in a module
  - `dir()`: names defined (at the top level) of the current file
  - `dir(module_name)`:names defined in the imported module
- Several *dunder* methods, plus user-defined names

```
import voti

print(dir(voti))
# ['Libretto', 'MAX_VOTO', 'Voto', 'random', 'voto_casuale', ...]

print(dir())
# ['voti', ...]
```

```
from voti import Voto, Libretto

print(dir(voti))
# NameError: name 'voti' is not defined.

print(dir())
# ['Libretto', 'Voto', ...]
```

A module should not contain executable statements

- The `import` statement runs the module file to create the definitions of the various names
- If there are any instructions outside the defined classes/functions, they will be executed, too…

- If `voti.py` contains

```
v = voto_casuale()
print(repr(v))
```

- Then, the `import voti` statement in `main.py` will cause
  - Defining a new top-level name (`v`)
  - Calling `voto_casuale()`
  - Printing the random vote
- All this should not happen!

Solving the problem

- It is useful to have some code *inside the module*
  - Usually, test code to verify that the module works correctly
  - Sometimes, a whole program (with its top-level code) may be used as a module for a larger problem
- We want to allow in-module code, but we don't want it to run, when imported

- The solution is to check if the file is the top-level one, or an imported one: __name__

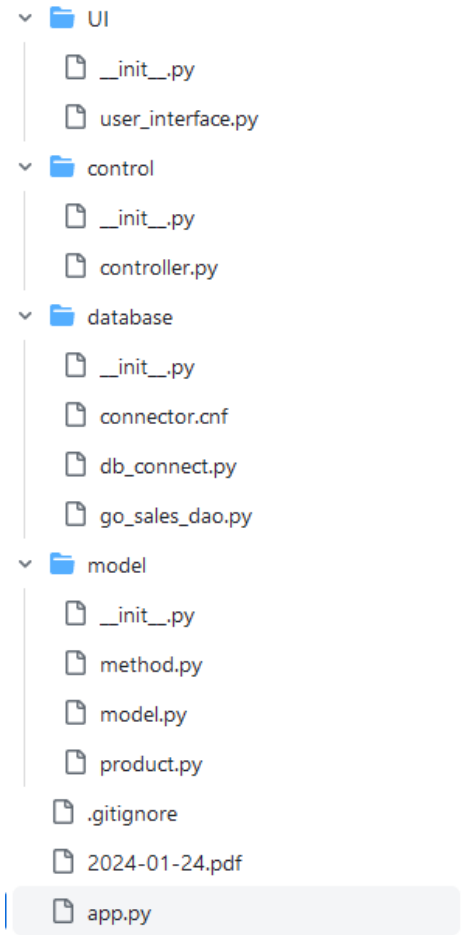
```
if __name__ == "__main__":
    v = voto_casuale()
    print(repr(v))
```

- Or, better:

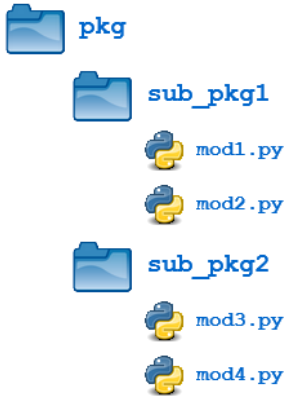
```
def _main():
    v = voto_casuale()
    print(repr(v))

if __name__ == "__main__":
    _main()
```

Packages

- When an application grows, it is no longer viable to have all the Python file in a single directory
- We can split groups of files in separate directories, called **packages**
  - Each directory is a package
  - The files of the directory are modules
  - They can be imported with the syntax `package_name.module_name`



(and sub-packages)



Importing from packages

- The traditional syntax still applies
  - import pkg.mod
  - from pgk.mod import name
  - from pgk.mod import name as alt_name
- Additionally, you may import modules from a package
  - from pkg import mod
  - from pkg import mod as alt_mod_name

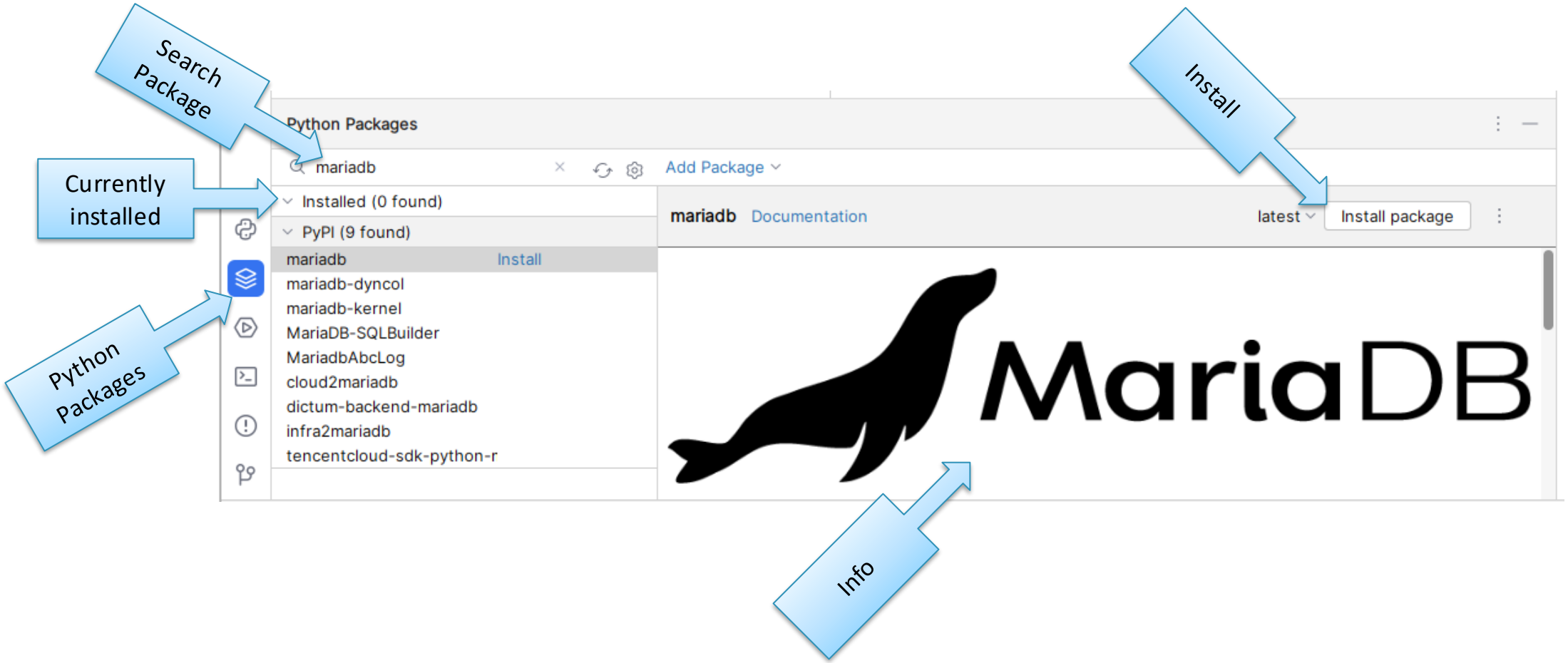
The `__init__.py` file

- Traditionally, the directory containing a package will also contain a special file
  - `__init__.py`
  - It was mandatory until Python 3.3, now it’s optional
- Can contain initialization statements, that are run when importing amy module from the package

Working with external modules

- To access a module from pypi.org, we must first install the module in our local Python interpreter
- Packages can be installed using the `pip` program
  - Search a project on https://pypi.org/
  - Install with `pip install project_name`
    - `pip install flet`
    - `pip install mariadb`
- Only installed packages can be imported

Installing packages with PyCharm

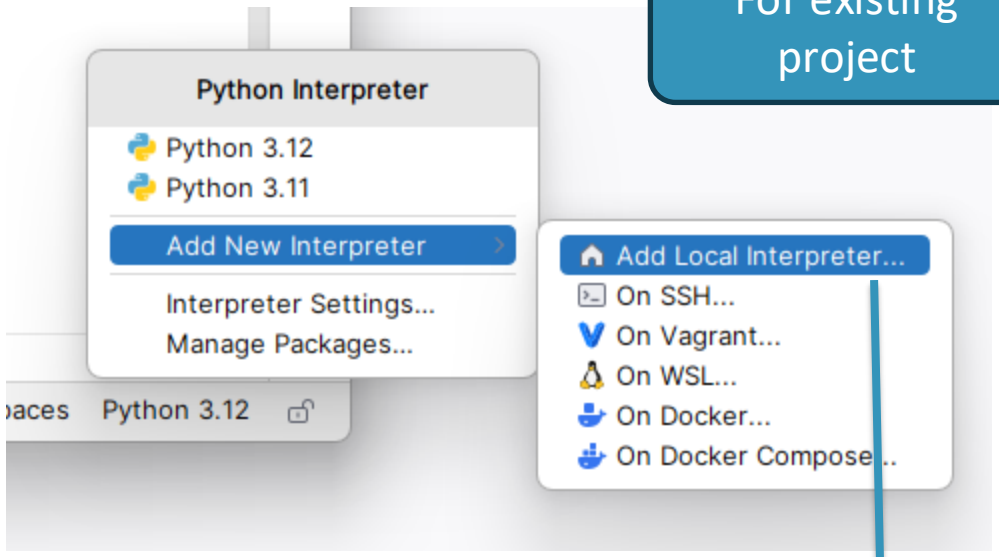


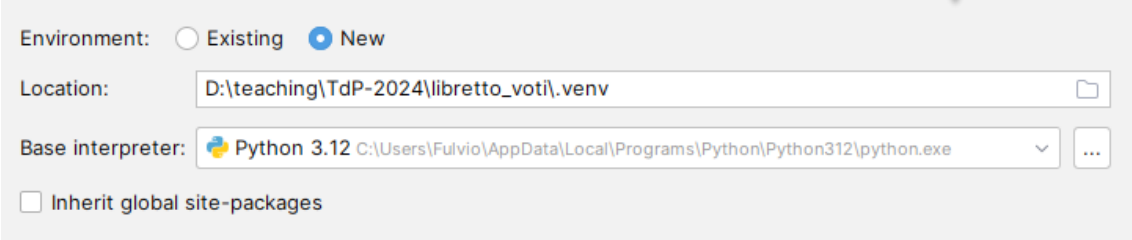
Virtual Environments

- Different projects may require different packages
- ⚠️ Your local Python library will contain all sorts of packages, that are used by some project
- ⚠️ When shipping a project, it's not clear which packages are needed to run it
- Python has a mechanism for separating the packages needed by each project
- 💡 Virtual environments
- It's a local "copy" of the Python interpreter, alongside with the packages needed for that project
- Stored in the `.venv` directory

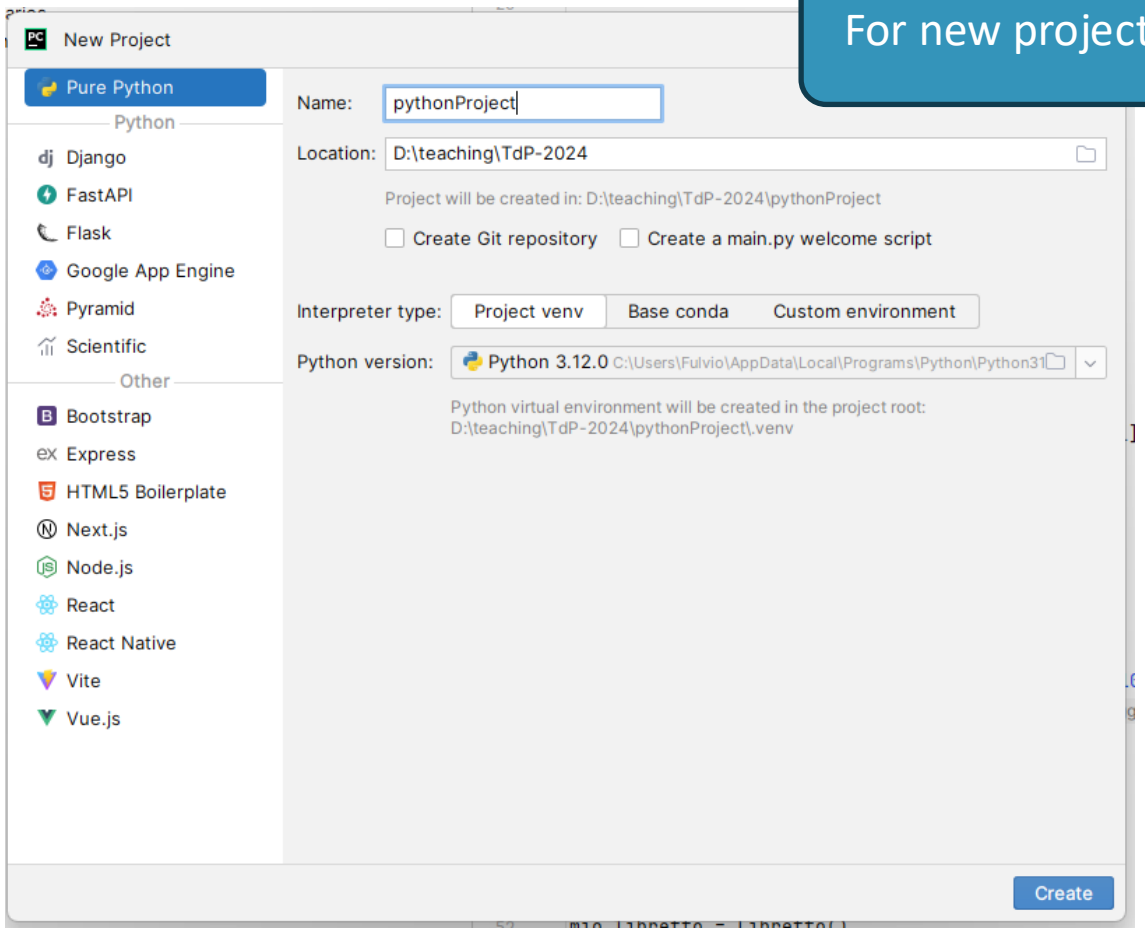
Virtual Environments in PyCharm

For existing project





For new project



Where does Python find packages?

- The import statement searches packages
  - In the current project directories
  - In the current virtual environment's library
  - In a set of directories defined by the Python installation

```
import sys
print(sys.path)
```

```
['C:\\Users\\Fulvio\\AppData\\Local\\Programs\\PyCharm
Professional\\plugins\\python\\helpers\\pydev',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\PyCharm
Professional\\plugins\\python\\helpers\\third_party\\thri
ftpy',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\PyCharm
Professional\\plugins\\python\\helpers\\pydev',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\PyCharm
Professional\\plugins\\python\\helpers\\pycharm_display',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\Python\\Pyt
hon312\\python312.zip',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\Python\\Pyt
hon312\\DLLs',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\Python\\Pyt
hon312\\Lib',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\Python\\Pyt
hon312', 'D:\\teaching\\TdP-2024\\libretto_voti\\.venv',
'D:\\teaching\\TdP-2024\\libretto_voti\\.venv\\Lib\\site-
packages',
'C:\\Users\\Fulvio\\AppData\\Local\\Programs\\PyCharm
Professional\\plugins\\python\\helpers\\pycharm_matplotli
b_backend', 'D:\\teaching\\TdP-2024\\libretto_voti']
```

requirements.txt

- A project may require several external packages
  - Installed with pip
  - Stored in the virtual environment
- How can we declare the information about the required packages?
  - So that other people may install them in their system
  - So that we can control which version numbers are installed

- Add a file `requirements.txt` to your project
  - Contains one line per package
  - May optionally specify the version number
  - PyCharm helps us synchronizing the file with the import statements

requirements.txt

```
pip==22.3.1
wheel==0.38.4
Pillow==9.5.0
setuptools==65.5.1
packaging==23.1
numpy==1.26.0
```

Don't specify version
Strong equality (==x.y.z)
Greater or equal (>=x.y.z)
Compatible version (~=x.y.z)

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